

Memo.

WTW158 Calculus

8

Class Test 1 (Mo)

4 - 8 March 2024

Time: 15 min

Surname:

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Initials:

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Student number:

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Group:

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

- (i) Consider $f(x) = \frac{x}{x^2 + x + 2}$. In the space below, write down what type of function f is.

Rational function (Algebraic $\frac{1}{2}$)

- (ii) Give an example of a reciprocal function.

see page 32

$$g(x) = \frac{1}{x^n}$$

✓ ① $n = 1, 2, 3, \dots$

$$g(x) = (x^2 - 1) + a^2$$

QUESTION 2 [2 marks]

Given: g is the function that assigns to the real numbers -1 and 1 , the value a^2 .

- (i) Give the algebraic version of the function g .

$$g(x) = a^2, x \in \{-1, 1\}$$

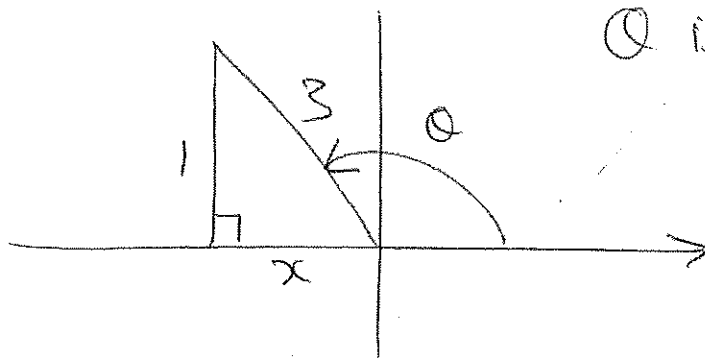
- (ii) Represent the function g in a table.

x	-1	1
$g(x)$	a^2	a^2

QUESTION 3 [2 marks]

Suppose $\sin \theta = \frac{1}{3}$ with $\theta \in (\frac{\pi}{2}, \pi)$. Find $\sec \theta$. Show all your steps.

Sketch
NB



θ is in the second quadrant.

Pythagoras:

$$x^2 + y^2 = r^2 \Rightarrow x^2 + 1^2 = 9 \Rightarrow x^2 = 9 - 1 = 8$$

$$x = -2\sqrt{2}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{-2\sqrt{2}}{3}} = \frac{-3}{2\sqrt{2}}$$

QUESTION 4 [2 marks]

Determine if $f(x) = \sin(2x) + |x|$ is even, odd or neither even nor odd. Show all your steps.

$$f(-x) = \sin(2(-x)) + |-x|$$

$$= \sin(-2x) + |x|$$

$$= -\sin(2x) + |x|$$

$$\neq \begin{cases} f(x) \\ -f(x) \end{cases}$$

$\Rightarrow f$ is neither odd nor even.

Memo.

WTW158 Calculus

Class Test 1 (Tuesday)

4 - 8 March 2024

Time: 15 min

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Group:

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Consider $f(x) = \frac{\sqrt{x}}{x^2 + x + 2}$. In the space below, write down what type of function f is.

Algebraic function

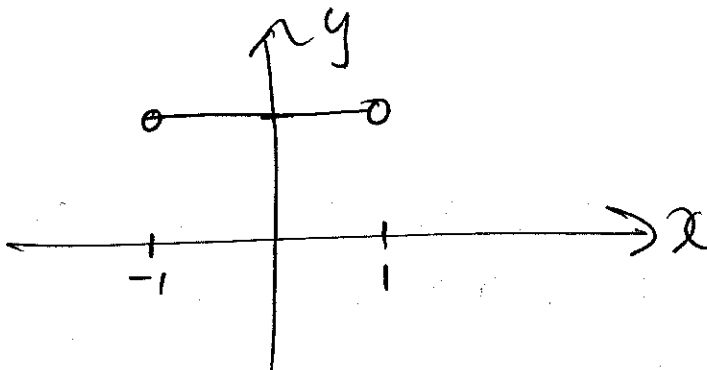
QUESTION 2 [2 marks]

Given: g is the function that assigns to each real number between -1 and 1 , the value a^2 .

(i) Give the algebraic version of the function g .

$g(x) = a^2, x \in (-1, 1)$

(ii) Sketch the graph of g . Indicate all relevant detail on the graph.



QUESTION 3 [1 mark]

Find the domain of the function $f(x) = \frac{2}{\sqrt{x-2}}$. Write your answer in interval notation.

$$\begin{aligned} x-2 &> 0 \\ \Rightarrow x &> 2 \\ \Rightarrow x &\in (2, \infty) \end{aligned}$$

QUESTION 4 [2 marks]

Suppose $\sin \theta = \frac{2}{3}$ with $\theta \in (\frac{\pi}{2}, \pi)$. Find $\sec \theta$. Show all your steps.

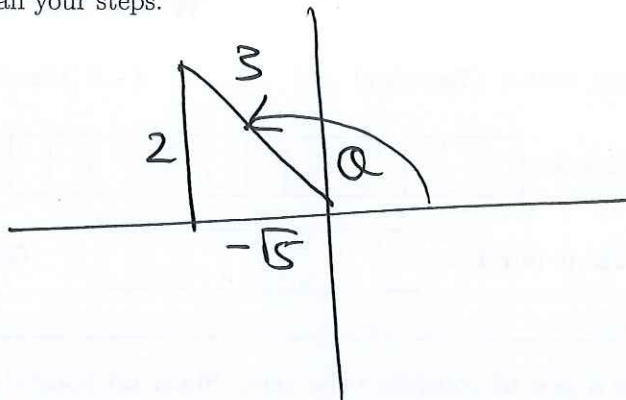
$$\sin \theta = \frac{2}{3}$$

$$x^2 + y^2 = r^2 \quad (\text{Pythagoras})$$

$$\Rightarrow x^2 + 4 = 9$$

$$x^2 = 9 - 4$$

$$= 5$$



θ in second quadrant

$$\Rightarrow x = -\sqrt{5}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{1}{\frac{-\sqrt{5}}{3}} = \frac{3}{-\sqrt{5}}$$

QUESTION 4 [2 marks]

Determine if the function $g(x) = |x^3| + x^4 + \cos(x^2)$ is even, odd or neither even nor odd.

$$g(-x) = |(-x)^3| + (-x)^4 + \cos(-x)^2$$

$$= |-x^3| + x^4 + \cos x^2$$

$$= |x^3| + x^4 + \cos x^2$$

$$= g(x)$$

$\Rightarrow g$ is an even function.

MB

WTW158 Calculus

Class Test 1 (Thursday)

4 - 8 March 2024

Time: 15 min

8

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

- (i) In the space below, write down a possible formula for a rational function with zeros 3 and 4, and with domain $(-\infty, 5) \cup (5, \infty)$.

$$\frac{(x-3)(x-4)}{(x-5)}$$

- (ii) Give an example of a root function defined on the real line.

$$\sqrt[3]{x} \quad (\text{see page 32})$$

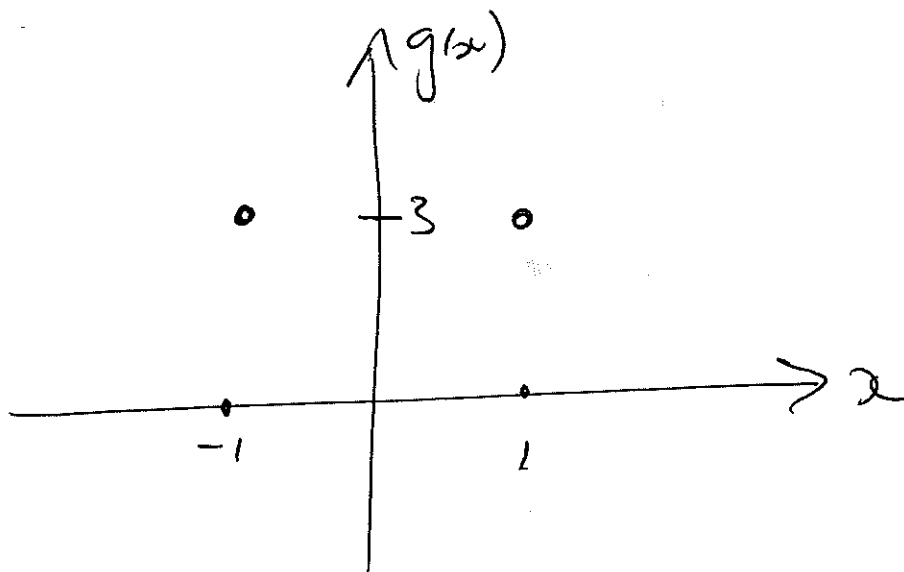
QUESTION 2 [2 marks]

Given: g is the function that assigns to the real numbers -1 and 1 , the value 3 .

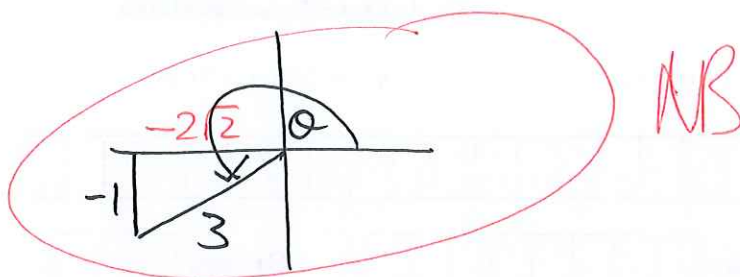
- (i) Give the algebraic version of the function g .

$$g(x) = 3, \quad x \in \{-1, 1\}$$

- (ii) Give the graphic representation of the function g .



Suppose $\sin \theta = -\frac{1}{3}$ with $\theta \in (\pi, \frac{3\pi}{2})$. Find $\cot \theta$. Show all your steps.



Pythagoras: $x^2 + y^2 = r^2$
 $x^2 + (-1)^2 = 9$
 $x^2 = 9 - 1 = 8 =$

Since θ in 3rd quadrant $x = -2\sqrt{2}$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{1}{\frac{-1}{-2\sqrt{2}}} = 2\sqrt{2}.$$

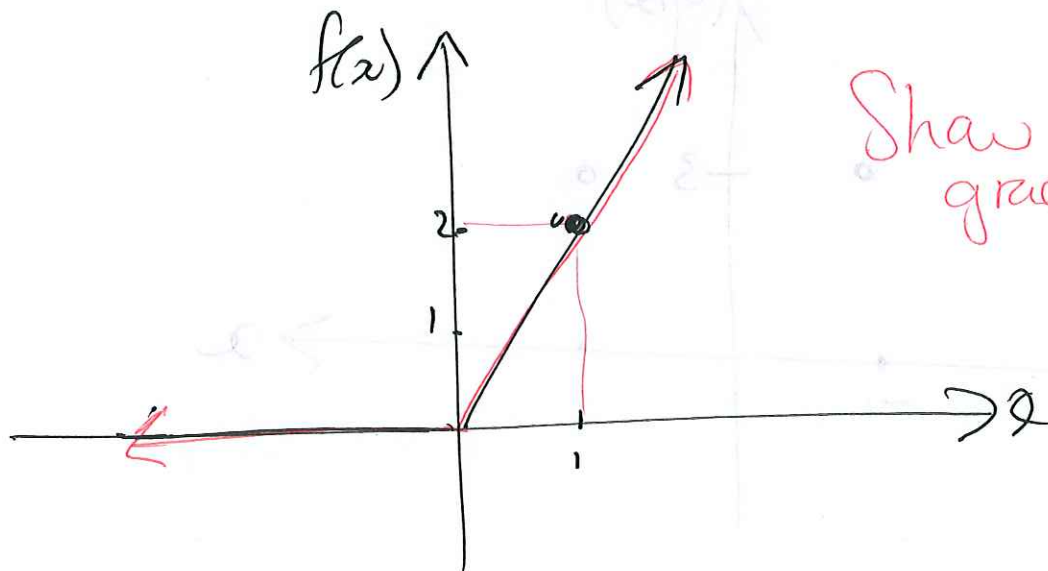
QUESTION 4 [2 marks]

Sketch the graph of the function $f(x) = |x| + x$. Show how you use the definition of the absolute value to simplify the function.

$$f(x) = |x| + x$$

$$= \begin{cases} x + x & \text{if } x \geq 0 \\ -x + x & \text{if } x < 0 \end{cases}$$

$$= \begin{cases} 2x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$



Show gradient

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Use a pen to complete the test. Sketches must also be in pen. Simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Consider the function $f(x) = \frac{x(x-2)}{\sqrt{x-2}}$. Classify function f by circling (all) the possible correct option(s) in the table below.

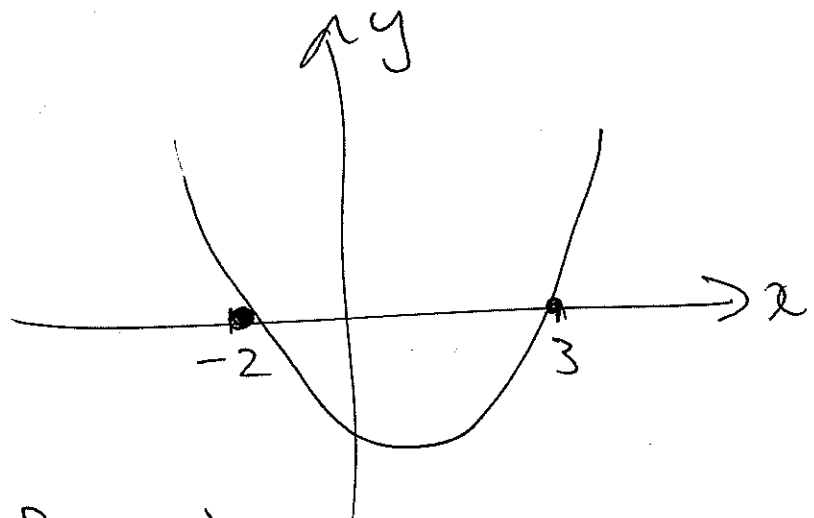
(a) Exponential function	☒ (b) Algebraic function	(c) Power function	(d) Rational function
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QUESTION 2 [2 marks]

Find the domain of the function $g(x) = \sqrt{x^2 - x - 6}$. Write your answer in interval notation.

$$x^2 - x - 6 \geq 0$$

$$\Rightarrow (x-3)(x+2) \geq 0$$



$$\Rightarrow x \in (-\infty, -2] \cup [3, \infty)$$

QUESTION 3 [2 marks]

Determine if the function $g(x) = |x - 3| + \cos x - 1$ is even, odd or neither even nor odd.

$$\begin{aligned} g(-x) &= |-x - 3| + \cos(-x) - 1 \\ &= |x + 3| + \cos x - 1 \end{aligned}$$

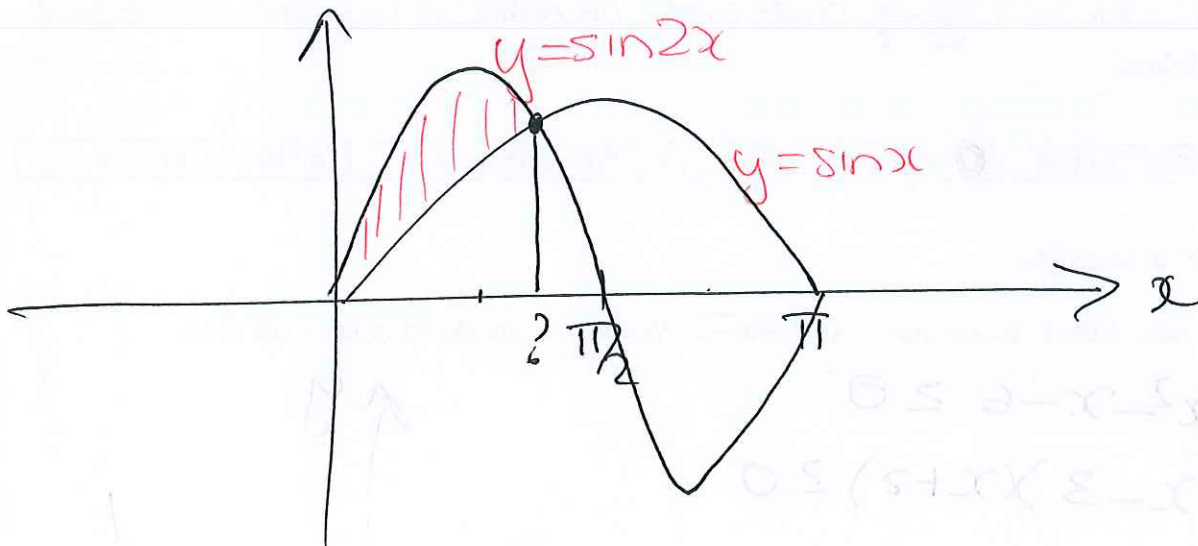
$$\neq \begin{cases} g(x) \\ -g(x) \end{cases}$$

g is neither even nor odd.

QUESTION 4 [3 marks]

Using a graph, find all values of x in the interval $[0, \pi]$ such that the inequality is satisfied.

$$\sin x < \sin 2x$$



$$\sin 2x = \sin x$$

$$\Rightarrow 2 \sin x \cos x = \sin x$$

$$\Rightarrow \cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

$$\Rightarrow \sin x < \sin 2x \text{ on } \left(0, \frac{\pi}{3}\right).$$